

How does ai use water? Like how does it cool the data centers?

Posted by u/Decent-Emergency3866 • 0 points • 13 comments

Just a thought that I have

Comments

u/piratesofthecaridina • 1 points • 2026-01-18 21:28 UTC

Here is documentation that addresses how water is used in data center cooling.

For a concise overview of how data centers are cooled, including air and liquid-based approaches, see:

- Wikipedia: Data center cooling https://en.wikipedia.org/wiki/Data_center_cooling

For detailed technical descriptions of cooling mechanisms used in data centers, including thermal management standards, see:

- ASHRAE TC 9.9 Power White Paper

https://www.ashrae.org/file%20library/technical%20resources/bookstore/ashrae_tc0909_power_white_paper_22_june_2016_revised.pdf

u/Yamabikio • 1 points • 2026-01-18 19:45 UTC

https://youtu.be/H_c6MWk7PQc

u/SlayerUnderSilence • 2 points • 2026-01-18 17:52 UTC

data centers have technology that gets hot as heck, and so they have water cooling systems that send water around to lower the temp to prevent stuff from stopping working or heck even exploding if it overheats. each time u give a prompt, the ai uses more energy = more heat = more water to counter that

and it only uses fresh water which is already scarce enough for humans

u/FlashyNeedleworker66 • 5 points • 2026-01-18 18:00 UTC

Scarce in some places.

Fresh water is a distribution problem, not a supply problem.

There's a reason you hear about the same data center in TN over and over again. We have a shit ton in the DMV and our water is fine. Funny how good infrastructure will do that.

u/MysteriousPepper8908 • 5 points • 2026-01-18 17:51 UTC

Pretty much how it does in a water-cooled PC, by running water next to the components to carry away the radiant heat.

u/DownWithMatt • 1 points • 2026-01-18 18:34 UTC

More or less, but from my understanding, current data centers are generally not a closed loop, but entirely evaporative, which I think is why the water consumption is a bigger concern.

The water is "consumed" and not just cooled and recycled.

I mean, it just technically puts the water back into the water cycle as it evaporates, so the water still "exists" by it is not longer available within the aquifer for local use.

u/ruk999 • 1 points • 2026-01-18 19:20 UTC

Some water evaporates and it's just returned to the water table it's not lost, it returns as rain, goes back in to rivers and such.

They also cycle water because impurities don't evaporate so the water gets more and more impure as water evaporates out and other water is added. So instead, they just cycle it out after a bit to keep the impurities down.

Some data centers just send it back to the sewer as waste. Not that while the water has more concentrated impurities, it didn't add any impurities that weren't already there. It just concentrated them in to waste water and purified water vapor.

Other data centers owned by big corps like Microsoft are doing purification themselves, so they're actually taking and removing impurities from the water they intake.

u/MysteriousPepper8908 • 2 points • 2026-01-18 18:41 UTC

Yes, that's right, the cooling itself is very similar but it's not typically a closed loop at these data centers. Closed loops do exist at data centers but it's more expensive and it's not required so there isn't much incentive to do it.

u/DownWithMatt • 0 points • 2026-01-18 18:45 UTC

So in other words, requiring it and enforcing it should be a no-brainer regulation? But of course we won't, because capitalism incentivizes profit at all cost, and our governance is entirely captured by lobbyists who would sacrifice entire swaths of humanity itself if it meant their quarterly earnings report increased by half a percentage point.

Awesome.

u/MysteriousPepper8908 • 1 points • 2026-01-18 18:50 UTC

Yeah, more or less. I still think it's a beneficial technology but are there reasonable regulations that should be imposed on where and how data centers are built but aren't because the administration thinks anything which limits corporations ability to do what they want are woke liberal politics? Definitely, hopefully we can move in that direction in 3 years but that's where we're at right now.

u/DownWithMatt • 2 points • 2026-01-18 18:59 UTC

Oh yeah, I agree. I think the technology itself is fantastic.

For example, Claude code, fucking game changer.

Yeah, most of the issues people have with AI itself is unrelated to the technology and is entirely the result of our current capitalist economic model.

Slop was not invented by AI, movies have been slop for decades, repackaging the same goddamn IP over and over with reboots and sequels. Video games have been mostly slop ever since the battle pass debuted. Music has been mass-produced slop for decades.

AI just lowered the barrier of entry for its creation, and capitalism incentivizes its production.

If you ask me, slop is defined or not as how something was produced, but why it was produced.

If it was produced for personal expression, for communication of one's internal mind state, it is art regardless if it was created using a brush, chisel, statistical model, or smearing your own feces. If it was produced for profit or primarily to generate income, it's slop. Full stop.

u/MysteriousPepper8908 • 1 points •
2026-01-18 19:33 UTC

I largely agree with that "primarily" distinction. If you want to devote your full energy to art, you aren't going to want to spend the majority of your time doing something else to survive so I don't think it's wrong to accept money for it but if that is your primary motivator then it does undermine your motivations.

u/DownWithMatt

• 1
points •
2026-
01-18
20:17
UTC

Right,
I
should
be
more
clear
when I
code-
switch
between
individual
and
systemic
framings.

What I
meant
is
moreso
how
"artwork"
produced
by
mega
corps
loses
their
actual
artistic
expression.

A
yearly
release
of a
call of
duty
game,
produced
on an
annual
cycle
as a
predictable
revenue
stream
is
devoid
of real

artistic
expression,
regardless
of how
many
individual
"artists"
it took
to
produce.

The
word
art
itself
becomes
meaningless
when
looking
at it
from a
corporate
perspective.

The
goal of
the
work
ceases
to be
an
expression
of the
human
"soul"
and
becomes
a line
item
on a
balance
sheet.
It
ceases
to
"artwork,"
to be
engaged
with,
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becomes
something
else
entirely,
content
to be
consumed.